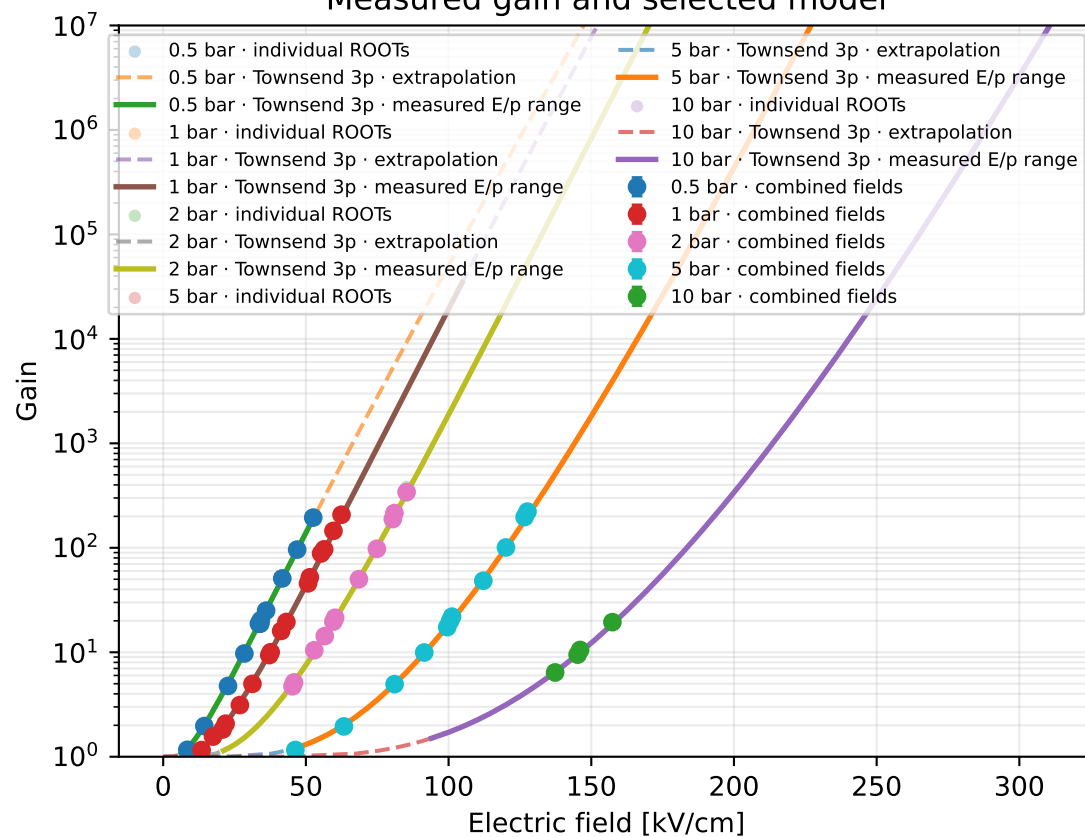
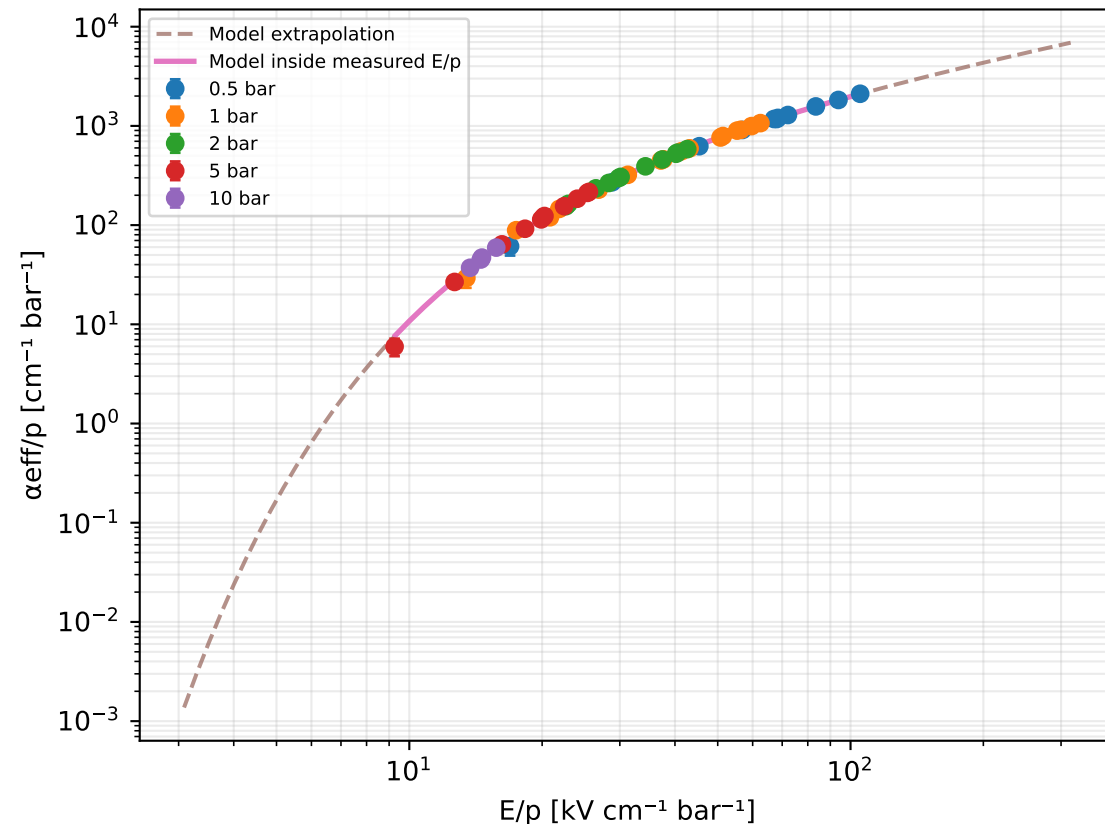


ArCF4 · Ar 99.5 % / CF4 0.5 % · gap 0.05 mm

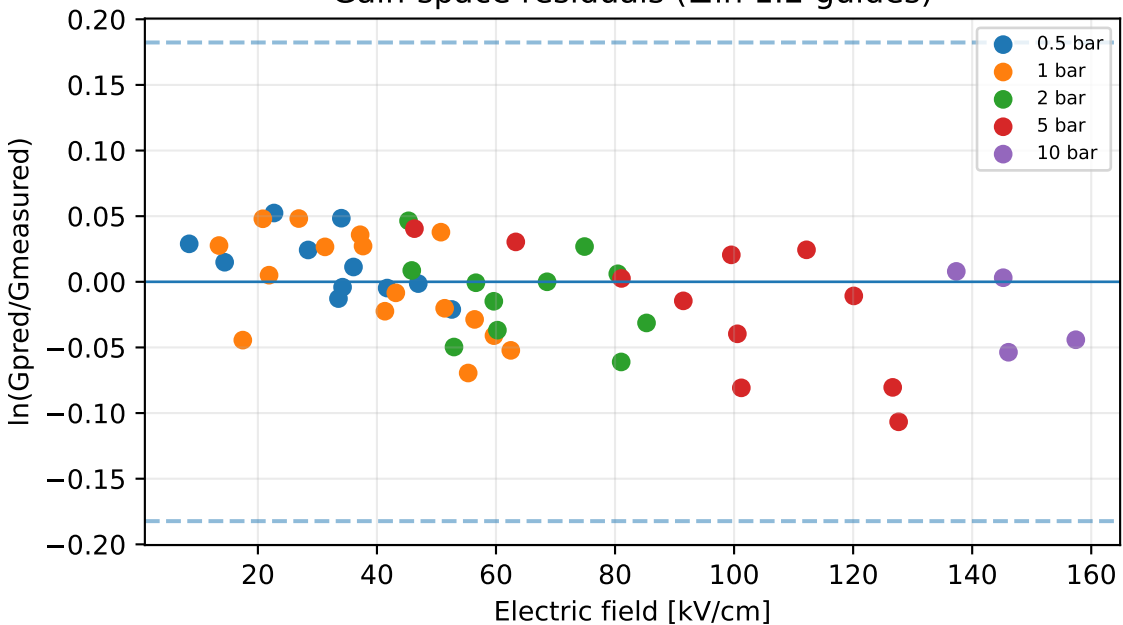
Measured gain and selected model



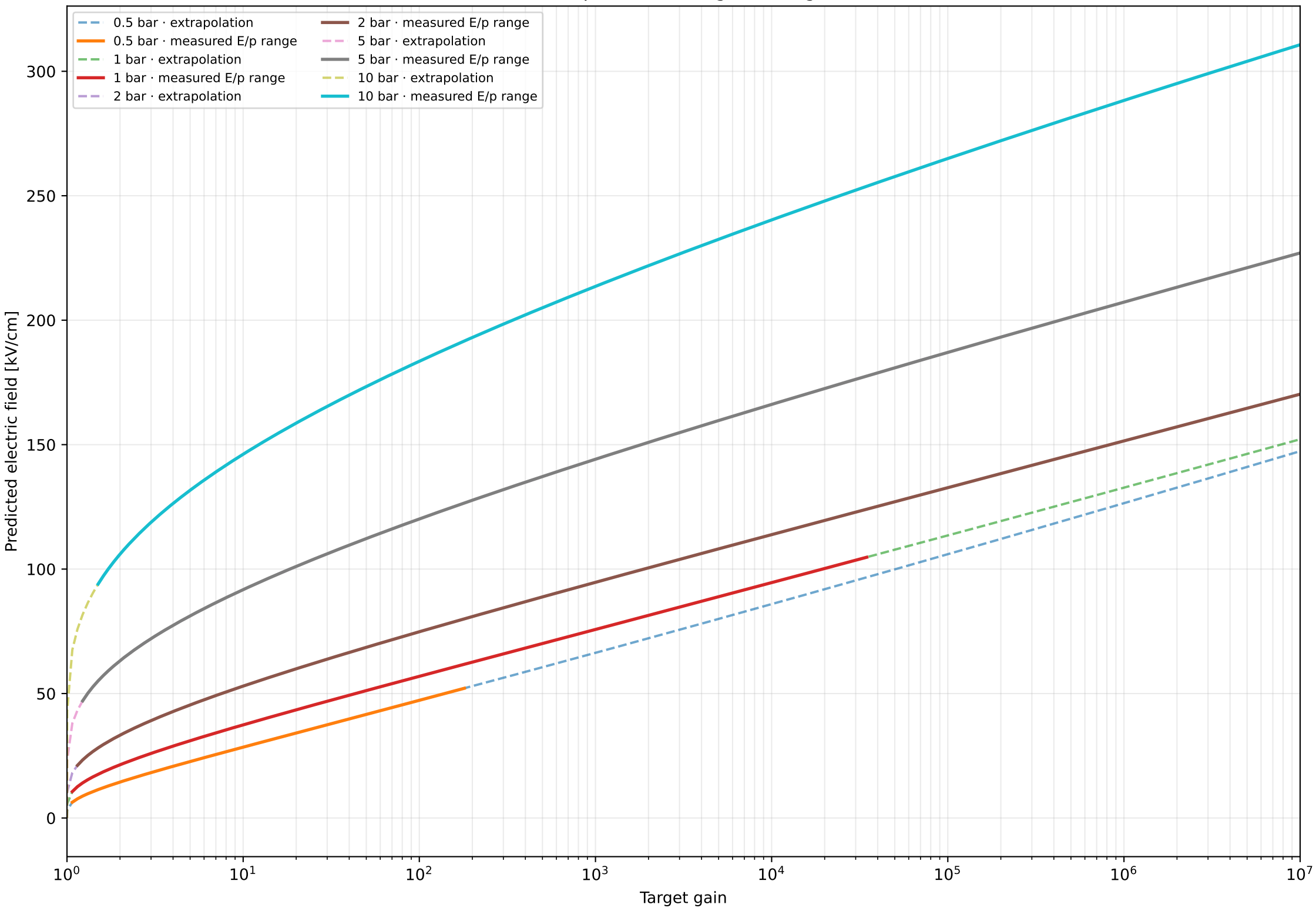
Reduced Townsend fit



Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 85
 Unique physical points: 53
 Pressures: 0.5, 1, 2, 5, 10 bar
 Selected model: Townsend 3p
 Status: accepted
 Training RMSE: $\times 1.054$
 Cross-validation median: $\times 1.028$
 Cross-validation worst: $\times 1.088$
 Reduced-field range: 9.26–105.1 $\text{kV cm}^{-1} \text{ bar}^{-1}$
 $A=49.368$, $B=35.433$, $m=0.87799$, $n=1$
 Covariance condition: 1.65e+04

Field extrapolation from gain 1 to gain 10^7 

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm ⁻¹]	npe	runs
0.5	8.4585	16.917	1.165468	0.0348	30.6245	139	1
0.5	14.4159	28.8318	1.965969	0.0589	135.197	382	1
0.5	22.7115	45.423	4.762419	0.143	312.151	463	1
0.5	28.4435	56.8869	9.736979	0.292	455.186	384	1
0.5	33.5356	67.0713	18.8118	0.563	586.897	356	1
0.5	34.0176	68.0352	18.77089	0.563	586.461	371	1
0.5	34.225	68.4499	20.2931	0.609	602.056	348	1
0.5	36.0587	72.1173	25.02312	0.266	643.96	3028	8
0.5	41.7151	83.4303	50.89773	1.52	785.964	352	1
0.5	46.9387	93.8774	96.10903	2.88	913.097	321	1
0.5	52.5728	105.146	194.5936	5.84	1054.18	283	1
1	13.4663	13.4663	1.157895	0.0347	29.3207	190	1
1	17.4658	17.4658	1.559486	0.0467	88.8712	311	1
1	20.8328	20.8328	1.822157	0.0546	120.004	343	1
1	21.8754	21.8754	2.071979	0.0621	145.701	389	1
1	26.8704	26.8704	3.124498	0.0936	227.855	498	1
1	31.272	31.272	4.986328	0.149	321.34	512	1
1	37.1772	37.1772	9.395393	0.282	448.044	521	1
1	37.665	37.665	10.01049	0.3	460.727	572	1
1	41.3391	41.3391	15.9958	0.479	554.465	476	1
1	43.1682	43.1682	19.50198	0.584	594.103	504	1
1	50.7482	50.7482	45.68103	1.37	764.337	464	1
1	51.3758	51.3758	52.19342	1.56	790.991	486	1
1	55.3278	55.3278	88.31	3.18	896.171	300	1
1	56.4097	56.4097	96.65275	2.9	914.225	455	1
1	59.6613	59.6613	145.2967	5.31	995.756	300	1
1	62.4876	62.4876	207.43	7.52	1066.96	300	1
2	45.3093	22.6547	4.726154	0.142	310.622	650	1
2	45.8506	22.9253	5.143928	0.154	327.563	667	1
2	52.945	26.4725	10.46142	0.314	469.539	635	1

p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
2	56.6227	28.3113	14.2918	0.429	531.937	634	1
2	59.6326	29.8163	19.66771	0.589	595.796	641	1
2	60.2369	30.1184	21.39587	0.641	612.64	629	1
2	68.5805	34.2902	50.0641	1.5	782.661	702	1
2	74.876	37.438	97.88833	3.13	916.765	600	1
2	80.461	40.2305	188.42	7.68	1047.73	300	1
2	81.0306	40.5153	215.12	9	1074.24	300	1
2	85.2886	42.6443	341.5807	4.49	1166.72	3000	10
5	46.2981	9.25962	1.16095	0.0348	29.8477	379	1
5	63.3243	12.6649	1.951311	0.0585	133.7	534	1
5	81.08	16.216	4.951705	0.148	319.946	704	1
5	91.4689	18.2938	9.949153	0.298	459.497	826	1
5	99.5338	19.9068	17.39277	0.522	571.211	802	1
5	100.548	20.1097	19.97613	0.599	598.908	838	1
5	101.217	20.2433	21.92655	0.657	617.54	885	1
5	112.164	22.4328	48.39172	1.45	775.866	725	1
5	120.108	24.0215	100.7483	3.3	922.525	600	1
5	126.647	25.3294	196.98	9.84	1056.62	300	1
5	127.64	25.528	221.9567	9.77	1080.5	300	1
10	137.329	13.7329	6.408748	0.0467	371.533	14860	17
10	145.213	14.5213	9.49571	0.285	450.168	1049	1
10	146.095	14.6095	10.51938	0.316	470.644	903	1
10	157.406	15.7406	19.4291	0.583	593.354	1072	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-193	×1.19	×2.62	yes	—
Townsend 3p	3	-306	×1.03	×1.09	yes	—
Townsend 4p	4	-315	×1.03	×1.08	yes	—